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ANIMAL PHYSIOLOGY

Effect of Chronic Administration of Aminoguanidine on Vascular Reactivity of the Greater Circulation in Rats with Monocrotaline-induced Pulmonary Hypertension

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Abstract—Pulmonary hypertension (PH) is a severe disease affecting both the pulmonary and systemic circulation. One of possible factors of these disturbances can be nitric oxide (NO) overproduction by inducible NO synthase (iNOS). To examine the effect of iNOS on systemic vascular reactivity, we used aminoguanidine (AG), a selective iNOS inhibitor. Using the model of monocrotaline-induced pulmonary hypertension, we demonstrated that chronic AG administration restores the decreased arterial pressure responses to NO donor and to nonspecific inhibitor of NO synthase as well as the decreased endothelium-dependent relaxation of isolated systemic artery. This points to an important role of iNOS in systemic pathogenesis of PH.

INTRODUCTION

Pulmonary hypertension (PH) can be described by a decreased pressure and vascular resistance in the pulmonary circulation, which leads to right ventricular hypertrophy. Animals with PH were used to study the main changes in the function of the pulmonary vascular system (Weir and Reeves, 1989; Hampl *et al*., 2000). PH proved to be a systemic disease affecting function of both the pulmonary circulation and other parts of the cardiovascular system (Comini *et al*., 1996; Chen *et al*., 2001). However, this problem remains insufficiently explored in experiments with animals.

High activity of the inducible nitric oxide synthase (iNOS) is observed in various types of PH (Takahashi *et al*., 1996; Takaya *et al*., 1998). Some publications report the relationship between high nitric oxide (NO) production in arterial hypertension and disturbed endothelium-dependent relaxation (EDR) of systemic vessels (Hong *et al*., 2000). In addition, high iNOS activity can result from inflammation accompanying progressing pulmonary hypertension (Lai *et al*., 1996); while high activity of this enzyme stimulating synthesis of both NO and other free radicals is observed in various disorders of the respiratory and cardiovascular systems (Kooy *et al*., 1995; Cosentino *et al*., 1998). Appearance of free radical molecules can affect function of proteins including enzymes playing the key role in EDR such as soluble guanylate cyclase (sGC) (Cherry *et al*., 1990). However, the problem of the role of iNOS in PH pathogenesis remains open.

The goal of this work was to study the function of the systemic circulation and the role of iNOS in PH pathogenesis using the model of monocrotaline-induced PH in rats. We used selective inhibitor of iNOS, aminoguanidine (AG), which was already used to examine the role of this enzyme in arterial hypertension (Hong *et al*., 2000).

MATERIALS AND METHODS

Experimental plan. The physiological model of PH was induced by subcutaneous injection of 60 mg/kg monocrotaline (MCT), an alkaloid of plant origin (Kaoru *et al*., 2000; Davydova *et al*., 2003) into rats weighing 180–200 g, while control rats were injected with saline. A week later, each group of rats was separated into two groups; the first one was daily administered with 15 mg/kg AG in drinking water, while the second group was administered with pure drinking water. Thus, the experiment included four groups of animals: control–control (C–C), control–AG (C–AG), MCT–control (MCT–C), and MCT–AG (MKT–AG). One month after MCT administration, the rats were used in a physiological experiment.

Catheter implantation. Plastic catheters (PE-10) were implanted into the right ventricle of the heart and femoral artery of rats under sodium pentobarbital anesthesia (40 mg/kg, ip). The right ventricular catheter was used only during the operation to monitor the right ventricular systolic pressure (RVSP), after which they were removed. The femoral arterial catheters were implanted and used in the subsequent chronic experiments (Davydova *et al*., 2003).

Measurement of the systemic and right ventricular pressures. RVSP was recorded for 20 min after a 20-min adaptation under anesthesia. Two days after the operation, the mean arterial pressure (MAP) and heart rate (HR) were determined in awake rats. The pressure and HR were recorded using an A/D converter (at a

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Table 1. Pathophysiological indications of development of pulmonary hypertension

Groups	Body weight, g	RVSP, mm Hg	Heart weight, mg	RVW, mg
C–C ($n = 8$)	298 ± 38	26 ± 3	890 ± 40	190 ± 10
C–AG ($n = 6$)	306 ± 31	27 ± 3	910 ± 40	200 ± 12
MCT–C ($n = 7$)	235 ± 39*	76 ± 13*	980 ± 55	330 ± 25*
MCT–AG ($n = 9$)	249 ± 24*	47 ± 8*#	980 ± 50	320 ± 30*

Notes: RVSP, right ventricular systolic pressure; RVW, right ventricular weight; C–C, C–AG, MCT–C, and MCT–AG, studied groups of animals.
* Differences from the corresponding control significant at $p < 0.05$.
\# Differences between MCT–AG and MCT–C groups significant at $p < 0.05$.

sampling rate of 512 Hz) and computer processing. MAP and HR were recorded after a 30-min adaptation in response to intravenous bolus injection of NO donor sodium nitroprusside (SNP) (4 μg/kg) and nonselective NO synthase inhibitor N-nitro-L-arginine methyl ester (L-NAME) (5 mg/kg). In addition to MAP and HR, we determined baroreflex sensitivity index (BI). BI was calculated by the formula: BI = $\Delta HR_{max}/\Delta MAP_{max}$, where ΔHR_{max} is the maximum change in HR induced by SNP injection and ΔMAP_{max} is the maximum change in MAP induced by injection of this compound (Davydova *et al.*, 2003).

Evaluation of right ventricular hypertrophy. At the end of the experiment, we weighed parts of the heart and recorded total heart weight, right ventricular weight (RVW), and left ventricular weight (LVW).

Experiments on isolated systemic vessels. The vessels of the greater circulation were isolated in decapitated animals on ice. Prior to isolation, the vessels were washed with cold saline to prevent thrombosis through a catheter inserted into the left femoral artery. A second-order artery (the superior muscular branch of the popliteal artery) was isolated with a small fraction of the preceding vessel with a larger diameter and fixed on a stainless steel needle with the inner diameter of 0.5 mm; other lower-order branches were tied off and left free. The length of the isolated vessels was 4 mm and the outer diameter of the artery was 300–400 μm.

The vessels were perfused at constant flow of 2 and 4 ml/min for the inner and outer ducts, respectively. The volume of the perfusion reservoir was 10 ml. Modified Krebs-Henseleit buffer saturated with 5% CO_2 and 95% O_2 and maintained at 38°C was used for perfusion (Nesterova *et al.*, 1999). The vascular response was evaluated by changes in perfusion pressure measured using a pressure transducer (Statham, United States). The digitized signal was recorded at 128 Hz sampling rate on a computer.

The constrictive tone (normally present in the body) was maintained by serotonin. In our experiment, the perfusion pressure increased to 100 mm Hg in response to 5×10^{-6} M serotonin. The vessel was perfused with acetylcholine (ACh) against the background of such contractile tone. EDR was induced by perfusion with ACh solutions at 1×10^{-8}, 5×10^{-8}, 1×10^{-7}, 5×10^{-7}, and 1×10^{-6} M. After recording ACh-induced EDR, the vessel was reperfused with serotonin to induce contractile tone (Davydova *et al.*, 2003).

Reagents. Aminoguanidine was provided by the Department of Animal Biochemistry at the Biological Faculty of the Moscow State University. Other reagents whose manufacturer was not mentioned were purchased from Sigma (United States).

Statistics. All values are given as mean ± SEM (SEM = $SD/\sqrt{n}$, where SD is standard deviation and n is the number arguments). The data were processed by the Mann-Whitney test using the Statistica 6.0 software. Differences were considered significant at $p < 0.05$.

RESULTS

In vivo experiments. PH developed in rats after MCT injection, as indicated by high RVSP and low body weight of the animals (Table 1), which agrees with published data (Mathew *et al.*, 1995; Kaoru *et al.*, 2000; Davydova *et al.*, 2003). Chronic administration of iNOS inhibitor, AG, decreased RVSP in MCT-induced PH, while RVSP remained high in the MCT–AG group relative to the control. AG administration had no effect on RVW and body weight in the MCT–AG animals as well (Table 1).

PH progression was manifested not only as high right ventricular pressure and hypertrophy but also as the change in the main indices and responses of the systemic circulation. For instance, MAP in animals with MCT-induced PH was significantly lower (102 ± 4 mm Hg in the MCT–C group) as compared to the control (116 ± 2% in the C–C group; $p < 0.05$) (Table 2). At the same time, HR did not significantly differ in PH and control animals. Moreover, LVW did not significantly differ in animals with MCT-induced PH and control ones.

Animals with MCT-induced PH showed a decreased blood pressure response to SNP administration in the MCT–C group (–17.5 ± 1.6 mm Hg) relative to the C–C group (–23.5 ± 1.5 mm Hg; $p < 0.05$) (Fig. 1a). An increase in MAP induced by L-NAME administration was +20.7 ± 4.2 mm Hg in the MCT–C group, which was significantly lower than in the C–C group (33.8 ± 2.0 mm Hg; $p < 0.05$) (Fig. 1b). Hence, blood pressure

Table 2. Indices of the function of systemic circulation

Groups	LVW, mg	MAP, mm Hg	HR, bpm	BI, mm Hg/min
C–C ($n = 8$)	460 ± 50	116 ± 2	362 ± 38	2.2 ± 0.6
C–AG ($n = 6$)	480 ± 40	117 ± 3	345 ± 34	2.2 ± 0.6
MCT–C ($n = 7$)	450 ± 50	102 ± 4*	347 ± 51	2.2 ± 0.7
MCT–AG ($n = 9$)	480 ± 50	102 ± 6*	386 ± 55	1.4 ± 0.6

Notes: LVW, left ventricular weight; MAP, mean arterial pressure; HR, heart rate; BI, baroreflex index calculated for blood pressure response to SNP.

response to NO donor and nonspecific inhibitor of NO synthase were significantly lower in the MCT–C group as compared to the control.

BRI calculated for blood pressure response to SNP administration did not differ in the MCT–C and C–C groups (Table 2).

Chronic AG administration restored some responses of the greater circulation in animals with MCT-induced PH. Blood pressure response to SNP and L-NAME administration was restored. For instance, blood pressure response to SNP administration increased in the MCT–AG group (–25.6 ± 1.6 mm Hg) relative to the MCT–C group ($p < 0.05$) and was similar to the control (Fig. 1a). L-NAME-induced increase in MAP proved to be significantly higher in the MCT–AG group (+34.6 ± 3.6 mm Hg) as compared to the MCT–C group (+20.7 ± 4.2 mm Hg; $p < 0.05$) and similar to that in the corresponding control group (+36.2 ± 5.7 mm Hg)

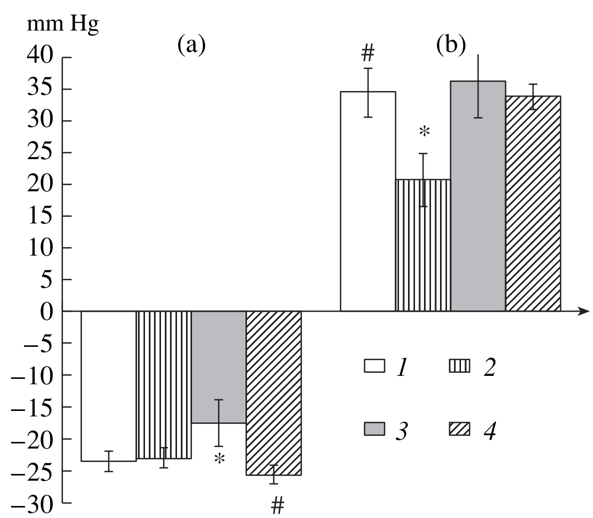

Fig. 1. Blood pressure response to intravenous administration of SNP (a) and L-NAME (b) in awake animals; *1*, C–C (n = 10); *2*, C–AG (n = 6); *3*, MCT–C (n = 10); *4*, MCT–AG (n = 8); ordinate: MAP changes, mm Hg; abscissa: groups of studied animals. (*) Differences from the C–C group significant at $p < 0.05$. (#) Differences between MCT–AG and MCT–C groups significant at $p < 0.05$ (for Figs. 1 and 2).

(Fig. 1b). At the same time, AG administration induced no changes in the response to SNP and L-NAME in control animals (Fig. 1). BI calculated for blood pressure response to SNP administration did not significantly differ between MCT–AG and C–AG groups (Table 2). Hence, chronic administration of iNOS inhibitor in MCT-induced PH restored normal blood pressure response to NO donor or nonspecific inhibitor of NO synthase.

At the same time, AG administration did not restore normal MAP. It remained significantly lower in the MCT–AG group (102 ± 6 mm Hg) as compared to the C–AG group (117 ± 3 mm Hg; $p < 0.05$). Similarly, AG administration had no effect on LVW or HR (Table 2).

Experiments on isolated systemic vessels. Assay for EDR of isolated systemic vessels with intact endothelium in a state of serotonin-induced tone demonstrated dose-dependent relaxation of pulmonary vessels induced by ACh perfusion in all four groups. ACh in a concentration range from 5×10^{-8} to 1×10^{-6} M induced significantly different systemic vascular relaxation in the animal groups. For instance, 5×10^{-7} M ACh induced a significantly lower vasodilatory response in the MCT–C group (–9 ± 3%) than in the C–C group (–23 ± 2%; $p < 0.05$). In the MCT–AG group, vascular relaxation was significantly higher for all ACh concentrations (–23 ± 6, –39 ± 9, –46 ± 9, –54 ± 10, and –60 ± 12%, respectively) as compared to both the MCT–C (–4 ± 2, –7 ± 2, –9 ± 3, –9 ± 3, and –12 ± 4%, respectively; $p < 0.05$) and C–AG groups (–11 ± 4, –15 ± 2, –26 ± 6, –29 ± 4, and –28 ± 3%, respectively; $p < 0.05$) (Fig. 2).

Hence, animals with MCT-induced PH demonstrated a decreased EDR in pulmonary vessels, while chronic AG administration significantly increased vasodilatory response in animals with PH to the levels exceeding the control ones.

DISCUSSION

The obtained data indicate that MCT-induced PH changes the functioning of both the pulmonary and systemic circulation. Animals with PH demonstrate significant decrease in MAP, blood pressure response to administration of hypertensive agent, and NO contribution to blood pressure control (experiments with nonspecific inhibitor of NO synthase). Our experiments demonstrated a decreased EDR in animals with MCT-induced PH as well.

Changes in the systemic circulation in MCT-induced PH are considered in a small number of publications. For instance, a decreased MAP was shown in MCT-induced PH. MCT-induced PH affects the mechanical function of the heart due to the development of right ventricular hypertrophy affecting activity of the heart as a whole (Chen *et al*., 2001), although we observed no significant changes in LVW. PH also disturbs the balance of vasoactive factors synthesized and metabolized in the lung (Weir and Reeves, 1989).

Previously we demonstrated that decreased EDR of the systemic vessels indicates a direct disturbance of the enzyme systems in the vascular endothelium and smooth muscle in the greater circulation, which are responsible for relaxation of these vessels (Davydova *et al*., 2003). These findings agree with the decreased eNOS content in the endothelium of aorta in rats with MCT-induced PH demonstrated by Comini *et al*., 1996). Disturbed functioning of the greater circulation could result from the factors that directly affected the pulmonary circulation, which could cause PH. Such factors can include inflammation accompanying progression of MCT-induced PH. Inflammation can play a significant role in pathogenesis of both the systemic hypertension (Kristal *et al*., 1998) and primary PH (Martin *et al*., 2002). At the early stages of PH, inflammation develops as a result of intoxication by an MCT metabolite including its effect on pulmonary vascular endothelium (Wilson *et al*., 1992). The number of leukocytes in the blood considerably increases 7 days after MCT injection and remains high up to day 28 (Kato *et al*., 1997). Miyata *et al*. (1995) demonstrated an increased production of various anti-inflammatory cytokines such as interleukin-1, interleukin-6, and tumor necrosis factor by alveolar macrophages in MCT-induced PH. Since inflammation accompanies PH, the model of MCT-induced PH is widely used as the most adequate model of PH (Mathew *et al*., 1995; Kaoru *et al*., 2000). Moreover, thrombosis, particularly, in the postcapillary circuit, and pulmonary phlebitis are typical for MCT-induced PH (Wilson *et al*., 1992), which makes the model similar to the pulmonary vein occlusion syndrome, which is a pathophysiological manifestation of the primary PH (Weir and Reeves, 1989).

Hence, inflammation can be a factor of low EDR of the systemic vessels in MCT-induced PH. High iNOS activity is observed in many cardiovascular diseases accompanied by inflammation (Hobbs *et al*., 1999). This enzyme can produce considerable quantities of NO or, in the absence of the substrate, superoxide anion (Hobbs *et al*., 1999; Packiasamy *et al*., 2003). High NO levels proved to decrease EDR of vessels in rats with systemic hypertension (Hong *et al*., 2000). NO and superoxide anion inhibit sGC activity (Cherry *et al*.,

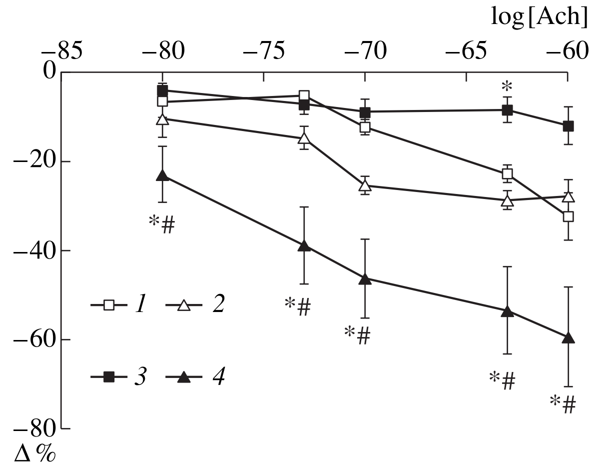

Fig. 2. Endothelium-dependent relaxation of the isolated popliteal artery in response to perfusion with acetylcholine solutions in concentrations from 1×10^{-8} to 1×10^{-6} M against the background of contractile tone induced by perfusion with serotonin solution; *1*, C–C ($n = 6$); *2*, C–AG ($n = 6$); *3*, MCT–C ($n = 7$); *4*, MCT–AG ($n = 6$); ordinate: relative perfusion pressure, Δ%; abscissa: common logarithm of acetylcholine (ACh) molar concentration.

1990; Fillipov *et al*., 1997). High iNOS activity is an important factor in pathogenesis of various types of hypertension. For instance, high iNOS activity and nitrate/nitrite concentrations were revealed in SHR rats with systemic hypertension (Hong *et al*., 2000) and in children with congenital ventricular septal defect (Takaya *et al*., 1998).

Previously we demonstrated a high iNOS activity in pulmonary vascular homogenates in rats with MCT-induced PH (Davydova *et al*., 2003). Published data indicate high plasma levels of nitrite/nitrate in MCT-induced PH (Takahashi *et al*., 1996). As was already mentioned, an increased number of leukocytes and anti-inflammatory cytokines are observed in the blood in MCT-induced PH. These findings suggest that PH-associated chronic inflammation in the pulmonary circulation can activate leukocytes in the greater circulation, while an increased production of free radical molecules in the greater circulation can affect endothelium and smooth muscle of the systemic vessels.

We used AG in the concentration providing for selective inhibition of iNOS (Bryk *et al*., 1999; Hong *et al*., 2000) to test possible involvement of high iNOS activity in PH pathogenesis.

In this work, we demonstrate that chronic administration of AG in the course of MCT-induced PH restores blood pressure responses to SNP administration as well as the contribution of NO to MAP control (experiments with L-NAME) and significantly increases EDR of the systemic vessels, which even outperforms the control EDR level. These data confirm previously proposed relationship between disturbed vascular reactivity and high iNOS activity in MCT-induced PH. Apparently, an increased production of free radical molecules resulting from the activation of leukocytes can affect functioning of endothelium-dependent systemic vascular relaxation mediated by sGC. This can also inhibit eNOS in the endothelium and sGC in the vascular smooth muscle. At the same time, AG removes the cause of their inhibition—increased production of NO and superoxide anion by leukocytes.

Explanation of excessive EDR levels in animals with MCT-induced PH is not so straightforward. Despite a decrease in RVSP, PH with all typical disturbances in the lesser circulation was observed in these animals. Disturbed balance of vasoactive factors as well as the processes of synthesis, activation, and degradation of many vasoactive factors taking place in the pulmonary bloodstream during PH (Weir and Reeves, 1989) can adaptively increase iNOS expression or activity in the greater circulation. For instance, synthesis and excretion of vasodilator prostacyclin is decreased while that of vasoconstrictor thromboxane is, conversely, increased in patients with primary and secondary PH (Galie *et al.*, 2003). However, this can be manifested only when the factor inhibiting the enzymes involved in vascular relaxation is removed, which is observed in the case of chronic AG administration. Changed regulation of the systemic circulation after AG administration is indirectly confirmed by the trends to increased HR and decreased BI in the MCT–AG group of animals.

Thus, the obtained data confirm possible involvement of iNOS in the inhibition of NO-dependent vasodilatory responses in the systemic circulation in monocrotaline-induced pulmonary hypertension.

REFERENCES

Bryk, R. and Wolff, D.J., Pharmacological Modulation of Nitric Oxide Synthesis by Mechanism-Based Inactivators and Related Inhibitors, *Pharmacol. Therap.*, 1999, vol. 84, pp. 157–178.

Chen, L., Gan, X.T., Haist, J.V., Feng, Q., Lu, X., Chakrabarti, S., and Karmazyn, M., Attenuation of Compensatory Right Ventricular Hypertrophy and Heart Failure Following Monocrotaline-Induced Pulmonary Vascular Injury by the Na^+–H^+ Exchange Inhibitor Cariporide, *J. Pharm. Exp. Ther.*, 2001, vol. 298, no. 2, pp. 469–476.

Cherry, P.D., Omar, H.A., Farrell, K.A., Stuart, J.S., and Wolin, M.S., Superoxide Anion Inhibits cGMP-Associated Bovine Pulmonary Arterial Relaxation, *Am. J. Physiol.*, 1990, vol. 259, pp. H1056–H1062.

Comini, L., Bachetti, T., Gaia, G., Pasini, E., Agnoletti, L., Pepi, P., Ceconi, C., Curello, S., and Ferrari, R., Aorta and Skeletal Muscle NO Synthase Expression in Experimental Heart Failure, *J. Mol. Cell. Cardiol.*, 1996, vol. 28, no. 11, pp. 2241–2248.

Cosentino, F., Patton, S., d'Uscio, L.V., Werner, E.R., Werner-Felmayer, G., Moreau, P., Malinski, T., and Luscher, T.F., Tetrahydrobiopterin Alters Superoxide and Nitric Oxide Release in Prehypertensive Rats, *J. Clin. Invest.*, 1998, vol. 101, pp. 1530–1537.

Davydova, M.P., Postnikov, A.B., D'yakonov, K.B., Lyubitskii, O.B., and Medvedeva, N.A., Involvement of Tetrahydrobiopterin in Local Change of Endothelium-Dependent Vasorelaxation in Pulmonary Hypertension, *Ross. Fiziol. Zh. im. Sechenova*, 2003, vol. 89, no. 12, pp. 1516–1522.

Filippov, G., Bloch, D.B., and Bloch, K.D., Nitric Oxide Decreases Stability of mRNAs Encoding Soluble Guanylate Cyclase Subunits in Rat Pulmonary Artery Smooth Muscle Cells, *J. Clin. Invest.*, 1997, vol. 100, pp. 942–948.

Galie, N., Manes, A., and Branzi, A., Prostanoids for Pulmonary Arterial Hypertension, *Am. J. Res. Med.*, 2003, vol. 2, no. 2, pp. 123–137.

Hampl, V. and Herget, J., Role of Nitric Oxide in the Pathogenesis of Chronic Pulmonary Hypertension, *Physiol. Rev.*, 2000, vol. 80, no. 4, pp. 1337–1372.

Hobbs, A.J., Higgs, A., and Moncada, S., Inhibition of Nitric Oxide Synthase as a Potential Therapeutic Target, *Ann. Rev. Pharmacol. Toxicol.*, 1999, vol. 39, pp. 191–220.

Hong, H.-J., Loh, S.-H., and Yen, M.-H., Suppression of the Development of Hypertension by the Inhibitor of iNOS, *Br. J. Pharmacol.*, 2000, vol. 131, pp. 631–637.

Ito, K.M., Sato, M., Ushijimo, K., Nakai, M., and Ito, K., Alterations of Endothelium and Smooth Muscle Function in Monocrotaline-Induced Pulmonary Hypertensive Arteries, *Am. J. Physiol. Heart Circ. Physiol.*, 2000, vol. 279, pp. H1786–H1795.

Kato, S., Ohnuma, N., Ohno, K., Takasaki, K., Okamoto, S., Asai, T., Okuda, M., Nakamoto, T., and Iizuka, M., Changes in Sequestered Leukocytes and Platelets in the Pulmonary Microvasculature of Rats with Monocrotaline-Induced Pulmonary Hypertension, *Int. J. Microcirc. Clin. Exp.*, 1997, vol. 17, no. 6, pp. 290–297.

Kooy, N.W., Royall, J.A., Ye, Y.Z., Kelly, D.R., and Beckman, J.S., Evidence for *in vivo* Peroxynitrite Production in Human Acute Lung Injury, *Am. J. Respir. Crit. Care Med.*, 1995, vol. 151, pp. 1250–1254.

Kristal, B., Shurtz-Swirski, R., Chezar, J., Manaster, J., Levy, R., Shapiro, G., Weissman, I., Shasha, S.M., and Sela, S., Participation of Peripheral Polymorphonuclear Leukocytes in the Oxidative Stress and Inflammation in Patients with Essential Hypertension, *Am. J. Hypertens.*, 1998, vol. 11, no. 8, pt. 1, pp. 921–928.

Lai, Y.L., Thacker, A.A., and Diana, J.N., Hypoxemia and Elevated Tachykinins in Rat Monocrotaline Pneumotoxicity, *Lung*, 1996, vol. 174, pp. 195–203.

Martin, I., Humbert, M., Marfaing-Koka, A., Capron, F., Wolf, M., Meyer, D., Simonneau, G., and Angles-Cano, E., Plasminogen Activation by Blood Monocytes and Alveolar Macrophages in Primary Pulmonary Hypertension, *Blood Coagul. Fibrinolysis*, 2002, vol. 13, no. 5, pp. 417–422.

Mathew, R. and Zeballos, G.A., Role of Nitric Oxide and Endotelin-1 in Monocrataline-Induced Pulmonary Hypertension in Rats, *Cardiovasc. Res.*, 1995, vol. 30, pp. 739–746.

Miyata, M., Sakuma, F., Yoshimura, A., Ishikawa, H., Nishimaki, T., and Kasukawa, R., Pulmonary Hypertension in Rats. 1. Role of Bromodeoxyuridine-Positive Mononuclear Cells and Alveolar Macrophages, *Int. Arch. Allergy Immunol.*, 1995, vol. 108, no. 3, pp. 281–286.

Nesterova, M.A., Chuiko, A.A., Sokolova, R.I., Medvedev, O.S., and Medvedeva, N.A., Evidence for an Antihypertensive Factor from the Adrenal Medulla of SHR Modulating

Neurogenic Vasoconstriction, *Acta Physiol. Scand.*, 1999, vol. 167, pp. 195–202.

Packiasamy, A.R., Hayashi, T., Iguchi, A., and Ignarro, L.J., Concomitant Production of Nitric Oxide and Superoxide in Human Macrophages, *Biochem. Biophys. Res. Commun.*, 2003, vol. 310, pp. 367–370.

Takahashi, T., Kanda, T., Inoue, M., Suzuki, T., Kobayashi, I., Kodama, K., and Nagai, R., A Selective Type V Phosphodiesterase Inhibitor, E4021, Protects the Development of Right Ventricular Overload and Medial Thickening of Pulmonary Arteries in a Rat Model of Pulmonary Hypertension, *Life Sci.*, 1996, vol. 59, pp. PL371–PL377.

Takaya, J., Teraguchi, M., Nogi, S., Ikemoto, Y., and Kobayashi, Y., Relation between Plasma Nitrate and Mean Pulmonary Arterial Pressure in Ventricular Septal Defect, *Arch. Dis. Child.*, 1998, vol. 79, pp. 498–501.

Weir, E.K. and Reeves, J.T., *Pulmonary Vascular Physiology and Pathophysiology*, New York: Marcel Dekker, 1989. Translated under the title *Fiziologiya i patofiziologiya legochnykh sosudov*, Moscow: Meditsina, 1995.

Wilson, D.W., Segall, H.J., Pan, L.C., and Lame, M.W., Mechanisms and Pathology of Monocrotaline Pulmonary Toxicity, *Critical. Rev. Toxicol.*, 1992, vol. 22, pp. 307–325.